

Introduction to Electronics Engineering

PART 1 — COMPLETE NOTES

Module 1 (10 hrs): Semiconductor Diodes and their Applications

For First-Year Students — written in simple, easy language
Course: Introduction to Electronics Engineering (IEE) · Course Code: EC 101

What this module covers (as per the syllabus)

- Introduction to Electronics Engineering
- PN junction diodes — characteristics and calculation of diode resistance
- Temperature dependence and diode capacitance
- Zener diode — operation and applications
- Diode as a rectifier (half-wave and full-wave)
- Diode clipper and clamper circuits

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- 1 Introduction to Electronics
- 2 Atomic Structure and Semiconductor Fundamentals
- 3 The PN Junction Diode
- 4 Zener Diode — Operation and Applications
- 5 Applications of the PN Junction & Diode as a Rectifier
- 6 Diode Clippers and Clampers
- 7 Solved Numerical Examples
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1. Introduction to Electronics

1.1 What is Electronics?

Electronics is the branch of science and engineering that deals with the study of the flow of electrons through vacuum, gas, or semiconductors. It involves the design and application of circuits that use these electron flows to perform specific functions.

An **electronic circuit** is a closed path formed by the interconnection of electronic components, through which electric current flows to perform a specific function.

1.2 Types of Electronics

- **Analog Electronics** — works with continuously varying signals (for example, a microphone's output).
- **Digital Electronics** — works with signals having only two levels (0 and 1).
- **Power Electronics** — deals with the control and conversion of electrical power.
- **Microelectronics** — very small circuits built on chips (integrated circuits).
- **Optoelectronics** — devices that work with both light and electricity (LEDs, solar cells).
- **Communication Electronics** — circuits used to transmit and receive information.

2. Atomic Structure and Semiconductor Fundamentals

2.1 Atomic Structure

An **atom** is the smallest particle of an element that retains the chemical properties of that element. Atomic structure refers to the structure of an atom comprising a **nucleus** (centre) in which **protons** (positively charged) and **neutrons** (neutral) are present. The negatively charged particles called **electrons** revolve around the nucleus.

- Each permitted orbit has a fixed amount of energy. The larger the orbit (radius), the greater is the energy of the electron.
- If an electron gains additional energy (heat, light, etc.), it moves to a higher orbit — the atom is then in a **state of excitation**.

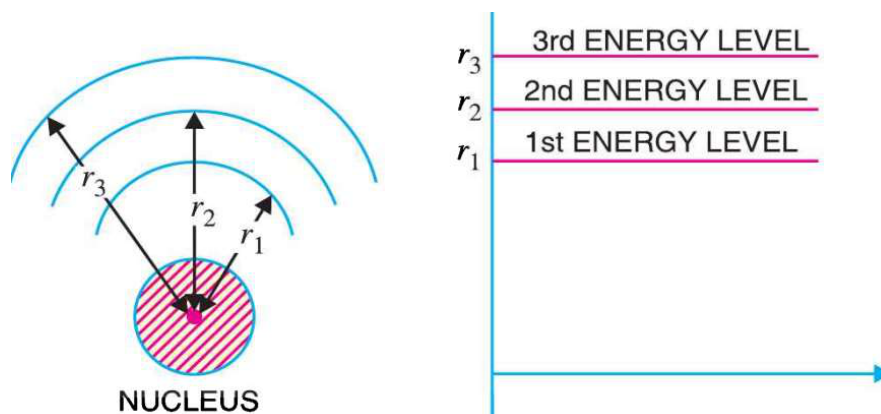


Figure 1: Bohr model of the atom — electrons revolve around the nucleus in fixed-energy orbits.

Key formulae

Atomic Weight = Number of Protons + Number of Neutrons

Atomic Number = Number of Protons = Number of Electrons in the atom

2.2 Semiconductor and Energy Bands

In terms of their electrical properties, materials are classified into three groups: **conductors**, **semiconductors**, and **insulators**.

Semiconductor: a material whose electrical conductivity lies between that of a conductor and an insulator, and whose conductivity can be controlled by adding impurities (doping) or by applying external fields. The behaviour of electrons in solids is governed by their arrangement in **energy bands**:

- **Conduction Band** — the energy band where electrons are free to move and contribute to electrical conductivity.
- **Valence Band** — the energy band where electrons are bound to atoms and do not contribute to conductivity.
- **Fermi Level** — the highest energy level that an electron can occupy at absolute zero temperature. It lies between the valence band and the conduction band.

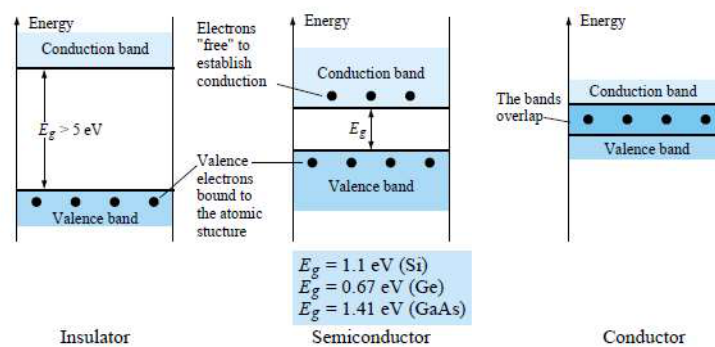


Figure 2: Energy band diagrams of an insulator, a semiconductor and a conductor — the forbidden energy gap decides the behaviour.

2.3 Comparison: Conductors, Semiconductors and Insulators

Parameter	Conductor	Semiconductor	Insulator
Forbidden Energy Gap	Not exist	Small (1 eV)	Large (> 5 eV)
Resistivity	Low	Moderate	High
Flow of current	Due to movement of free electrons	Due to movement of electrons and holes	Almost negligible; only free electrons
Charge carriers in conduction band	Completely filled	Partially filled	Completely vacant
Charge carriers in valence band	Almost vacant	Partially filled	Completely filled
Example	Copper, Aluminium, Graphite	Silicon, Germanium, Arsenic	Paper, Rubber, Glass, Plastic
Applications	Conducting wires, transformers, electrical cords	Diodes, transistors, optocouplers	Sports equipment, home appliances

2.4 Common Semiconductor Materials

Semiconductor materials fall into two classes: **single-crystal** and **compound**. The three semiconductors most frequently used in electronic devices are **Germanium (Ge)**, **Silicon (Si)**, and **Gallium Arsenide (GaAs)**.

In a pure silicon or germanium crystal, the four valence electrons of one atom form a bonding arrangement with four adjoining atoms — this is called **covalent bonding**.

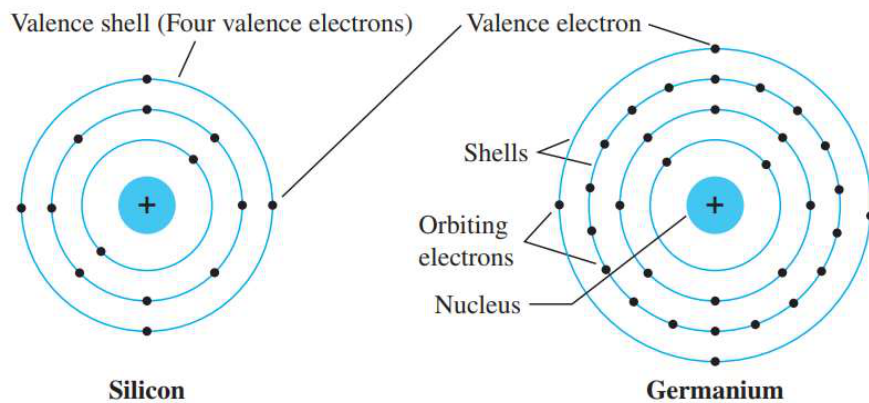


Figure 3: Atomic structure of silicon and germanium atoms — each has four valence electrons.

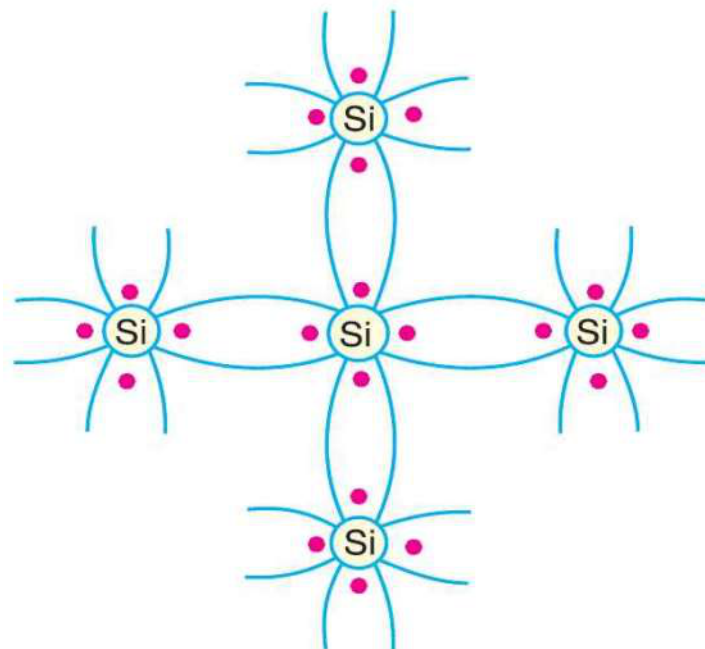


Figure 4: Covalent bonding in a pure silicon crystal — neighbouring atoms share valence electrons.

2.5 Types of Semiconductors

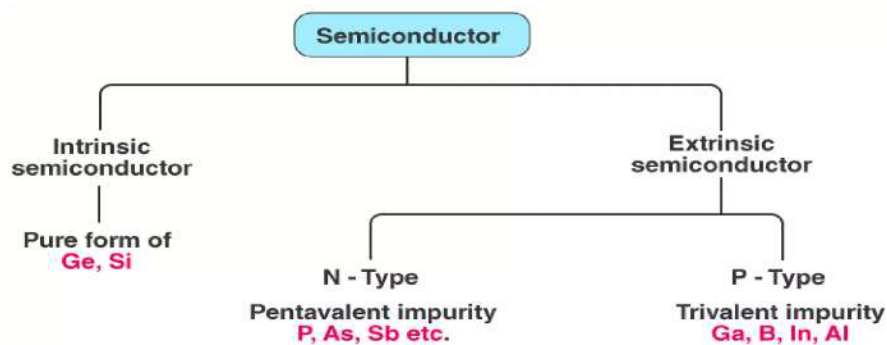


Figure 5: Classification of semiconductor materials.

A. Intrinsic Semiconductors

- Pure-form semiconductor materials without any significant impurities. Examples: Silicon (Si) and Germanium (Ge).
- The number of electrons in the conduction band equals the number of holes (empty states in the valence band).
- Conductivity is low at room temperature and increases with temperature.

B. Extrinsic Semiconductors (Doped)

Extrinsic semiconductors are created by adding specific impurities (called **dopants**) to an intrinsic semiconductor to enhance its electrical conductivity.

N-Type Semiconductor

- Created by adding pentavalent impurity elements (5 valence electrons) such as Antimony (Sb), Arsenic (As) and Phosphorus (P).
- The impurity atoms donate a free electron, so they are called **donor atoms**.
- Majority carriers: electrons. Minority carriers: holes.

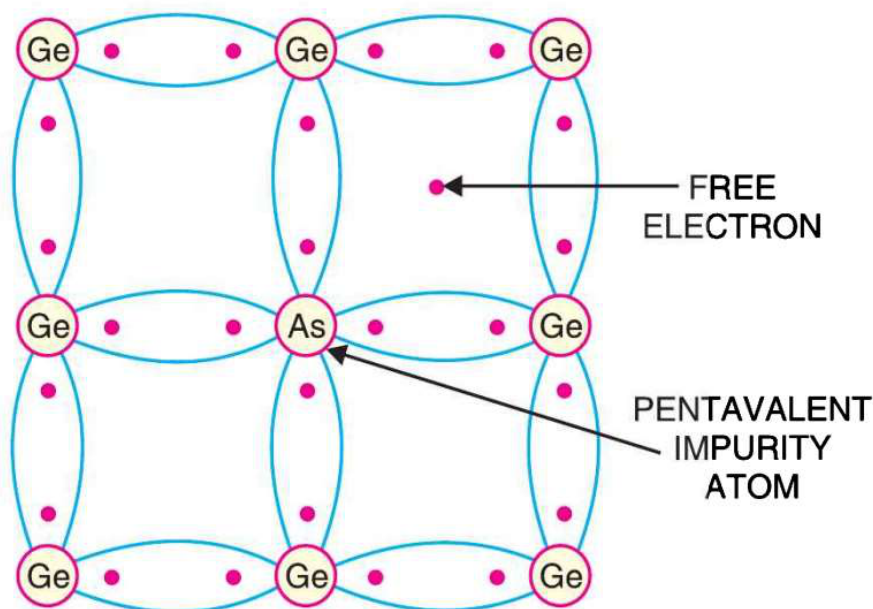


Figure 6: Arsenic (pentavalent) impurity in a crystal — the fifth electron becomes a free carrier (n-type material).

P-Type Semiconductor

- Created by adding trivalent impurity elements (3 valence electrons) such as Boron (B), Gallium (Ga) and Indium (In).
- The impurity atoms accept an electron, creating a hole; they are called **acceptor atoms**.
- Majority carriers: holes. Minority carriers: electrons.

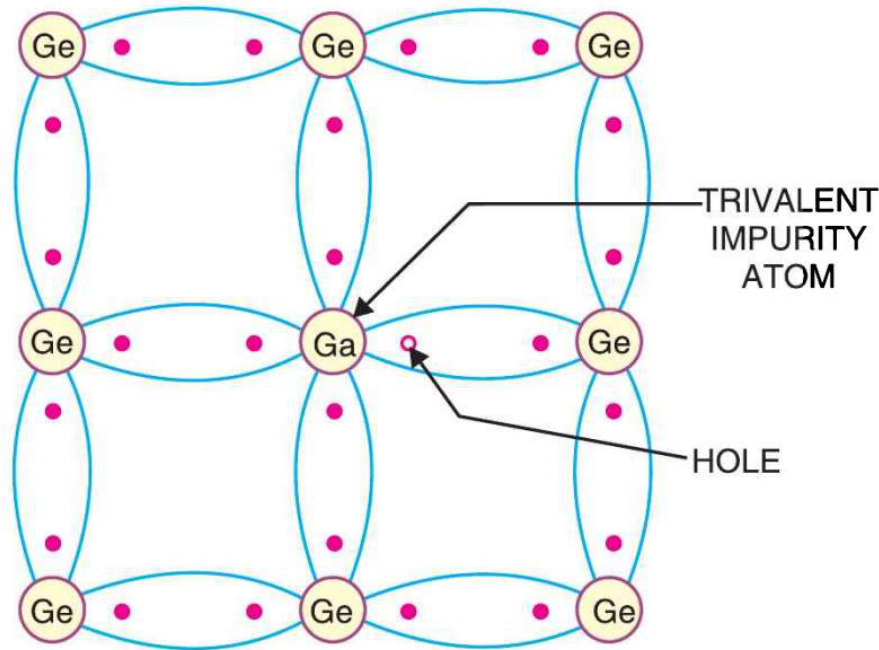


Figure 7: Gallium (trivalent) impurity in a crystal — the missing electron creates a hole (p-type material).

The controlled addition of dopants allows precise control over the electrical properties of semiconductors, making them essential in devices like diodes, transistors and integrated circuits.

2.6 Intrinsic vs. Extrinsic Semiconductors

Property	Intrinsic Semiconductor	Extrinsic (n-type & p-type)
Purity	Pure (no impurities)	Doped with impurities
Conductivity	Low at room temperature	Higher, due to the presence of free charge carriers
Charge carriers	Equal number of electrons and holes	Majority and minority carriers (electrons or holes)
Effect of temperature	Conductivity increases with temperature	Depends on doping and temperature
Types	Not applicable	n-type (electron majority) and p-type (hole majority)
Energy band structure	No additional energy levels	Donor or acceptor energy levels present

2.7 Majority and Minority Carriers

When a semiconductor is doped, one type of charge carrier becomes dominant:

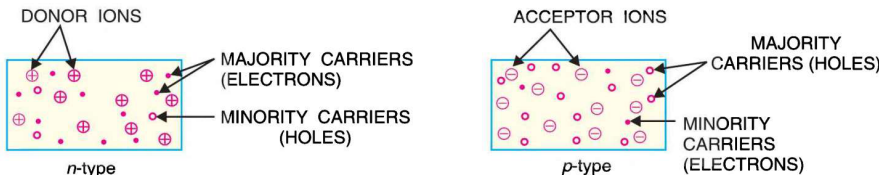


Figure 8: Majority and minority carriers in n-type and p-type materials.

Material Type	Majority Carrier	Minority Carrier
N-Type	Electrons	Holes

Material Type	Majority Carrier	Minority Carrier
P-Type	Holes	Electrons

3. The PN Junction Diode

3.1 Introduction

The PN junction is the **fundamental building block** of semiconductor devices such as diodes, transistors and integrated circuits. The PN junction diode is created by simply joining an n-type and a p-type semiconductor material together.

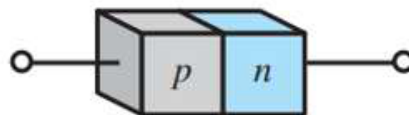


Figure 9: The PN junction diode — physical structure and circuit symbol.

A diode has three operating conditions:

- 1 **No bias** — no external voltage applied ($V_D = 0$).
- 2 **Forward bias** — positive terminal to the p-region, negative terminal to the n-region.
- 3 **Reverse bias** — positive terminal to the n-region, negative terminal to the p-region.

Bias means the application of an external voltage across the two terminals of a device to extract a response.

3.2 PN Junction with No Bias

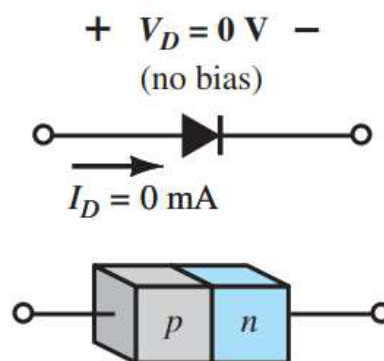


Figure 10: The unbiased PN junction.

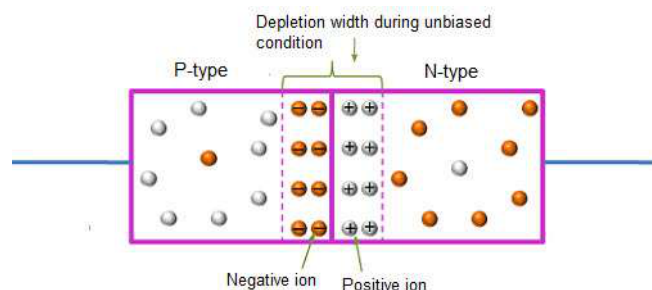


Figure 11: Formation of the depletion region at the junction.

- Electrons from the n-region diffuse into the p-region; holes from the p-region diffuse into the n-region. Movement continues until equilibrium is established.

- Electrons and holes recombine near the junction, creating a **depletion region** — a region with no free charge carriers. Only immobile ions remain (negative ions on the p-side, positive ions on the n-side).
- An electric field develops across the depletion region, opposing further diffusion.
- In the absence of applied bias, the net flow of charge in any direction is zero.

Characteristics of the depletion region

- **Width** — depends on doping concentration. Heavily doped regions have narrower depletion regions.
- **Charge distribution** — the p-region contains negatively charged acceptor ions; the n-region contains positively charged donor ions.
- **Electric field** — points from the positively charged n-region to the negatively charged p-region.

Barrier potential (built-in potential)

The electric field creates a potential barrier (V_b) that prevents further diffusion of charge carriers. At room temperature:

$$\text{Silicon: } V_b = 0.7 \text{ V} \quad \text{Germanium: } V_b = 0.3 \text{ V}$$

3.3 Reverse Bias

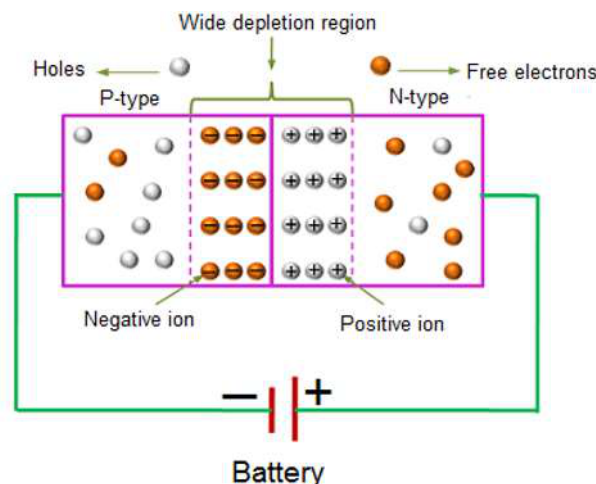


Figure 12: PN junction under reverse bias — the battery positive terminal connects to the n-region.

When the positive terminal of the supply is connected to the n-region and the negative terminal to the p-region:

- The potential barrier is increased.
- The depletion region widens.
- Charge carriers cannot cross the junction.
- Very little current flows (due to minority charge carriers only).
- If the reverse voltage exceeds a critical value, breakdown can occur through Zener or avalanche breakdown.

Reverse saturation current (I_s) — also called leakage current — is caused by thermally generated minority carriers swept across the junction by the built-in electric field. It is very small in magnitude, almost constant with reverse voltage, and increases strongly with temperature.

3.4 Forward Bias

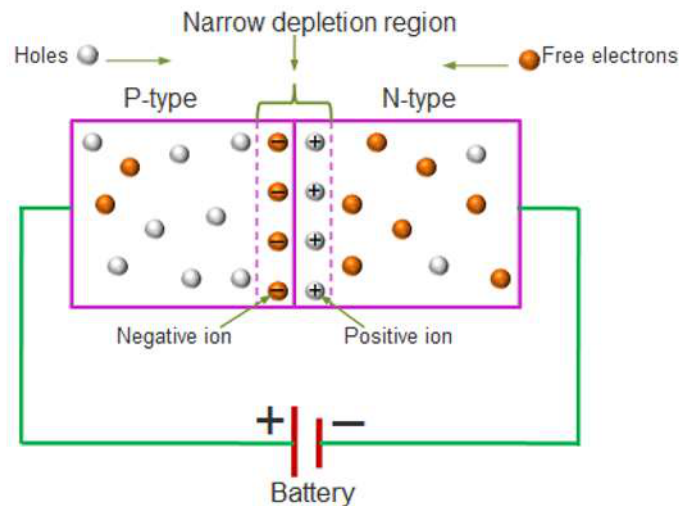


Figure 13: PN junction under forward bias — the battery positive terminal connects to the p-region.

When the positive terminal of the supply is connected to the p-region and the negative terminal to the n-region:

- The potential barrier is reduced.
- Charge carriers can cross the junction.
- Electrons move toward the p-region; holes move toward the n-region.
- A significant forward current flows.
- In silicon diodes, the forward voltage must be approximately 0.7 V for significant conduction.

3.5 Shockley's Diode Equation

The general current–voltage relationship of a semiconductor diode is given by **Shockley's equation**:

$$I_D = I_s(e^{V_D/nV_T} - 1) \quad (A)$$

Figure 14: Shockley's diode equation.

$$I_D = I_s (e^{V_D / n V_T} - 1)$$

$$V_T = \frac{kT_K}{q} \quad (V)$$

Figure 15: Thermal voltage formula.

k is Boltzmann's constant = 1.38×10^{-23} J/K

T_K is the absolute temperature in kelvins = $273 +$ the temperature in $^{\circ}\text{C}$

q is the magnitude of electronic charge = 1.6×10^{-19} C

Figure 16: Physical constants used with the diode equation.

Symbol	Definition
I_D	Diode current

Symbol	Definition
I_s	Reverse saturation current
V_D	Applied forward-bias voltage across the diode
n	Ideality factor: $n = 1$ for Germanium; $n = 2$ for Silicon
V_T	Thermal voltage = $k T_K / q$
k	Boltzmann's constant = 1.38×10^{-23} J/K
T_K	Absolute temperature in kelvins = $273 + \text{temperature } (^{\circ}\text{C})$
q	Magnitude of electronic charge = 1.6×10^{-19} C

3.6 Current–Voltage (I–V) Characteristics

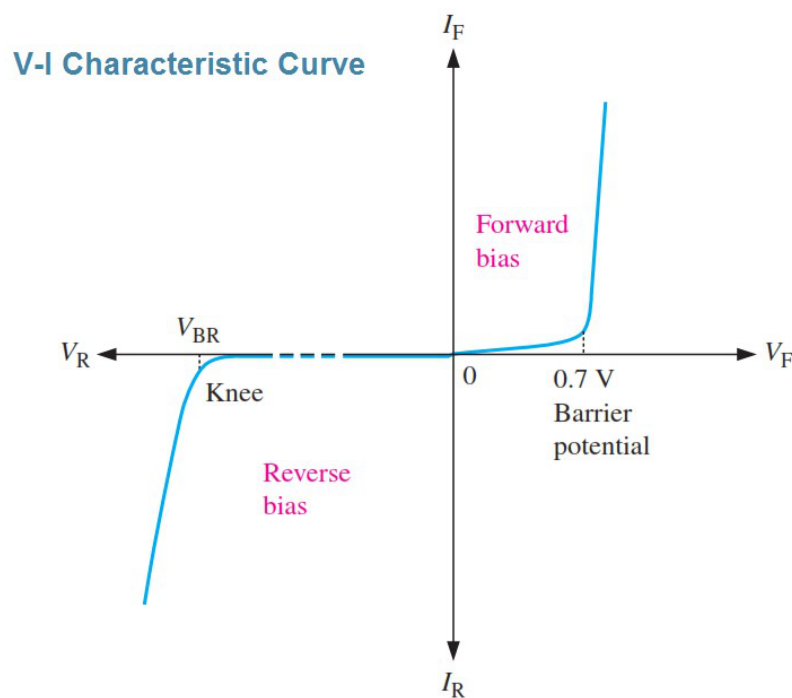


Figure 17: The complete I–V characteristic curve of a PN junction diode.

Forward bias region

- When the voltage exceeds the threshold (0.7 V for silicon, 0.3 V for germanium), current increases exponentially.
- The diode conducts current easily once the threshold is crossed.

Reverse bias region

- Only a very small reverse saturation current (I_s) flows.
- If the reverse voltage exceeds the breakdown voltage, the diode conducts large currents — this may permanently damage regular diodes.

Knee voltage is the minimum forward voltage after which the forward current starts increasing exponentially. For Si it is about 0.7 V; for Ge about 0.3 V.

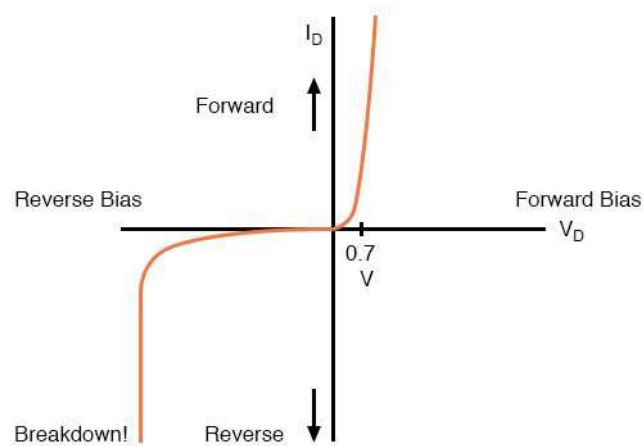


Figure 18: I–V characteristic showing the knee voltage and the sharp rise of current beyond it.

3.7 Diode Resistance

A. Static Resistance (DC Resistance), R_D

Defined at a specific operating point on the I–V characteristics:

$$R_D = V_D / I_D$$

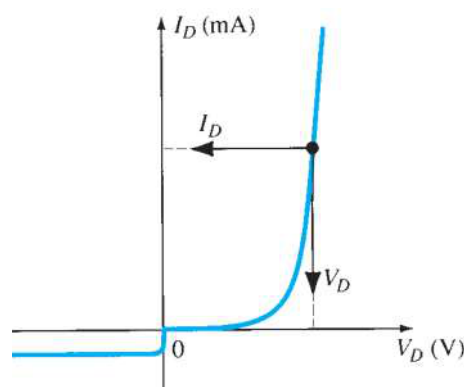


Figure 19: Static (DC) resistance is defined at one operating point on the curve.

B. Dynamic Resistance (AC Resistance), r_d

Calculated using the slope of the I–V curve at the operating point:

$$r_d = (\text{change in } V_D) / (\text{change in } I_D) = n V_T / I_D$$

At room temperature (27°C), V_T is about 26 mV, and for $n = 1$:

$$r_d = 26 \text{ mV} / I_D$$

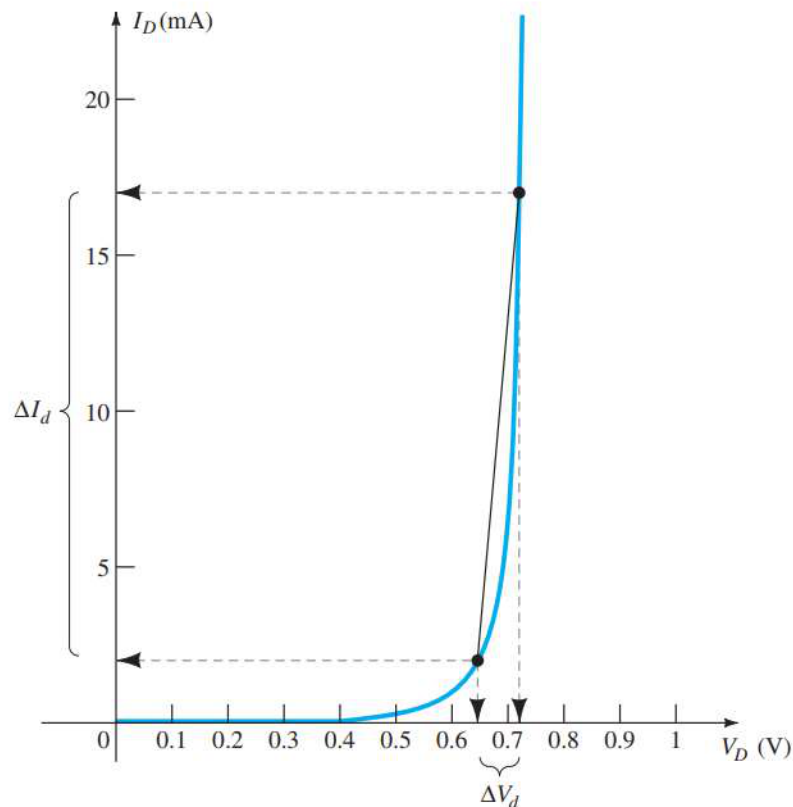


Figure 20: Dynamic (AC) resistance is found from the slope of the curve around the operating point.

3.8 Temperature Effects on Diodes

- As temperature increases, it adds energy to the diode.
- It reduces the required forward bias voltage for conduction.
- It increases the amount of reverse saturation current.
- Germanium diodes are more sensitive to temperature variations than silicon or gallium arsenide diodes.

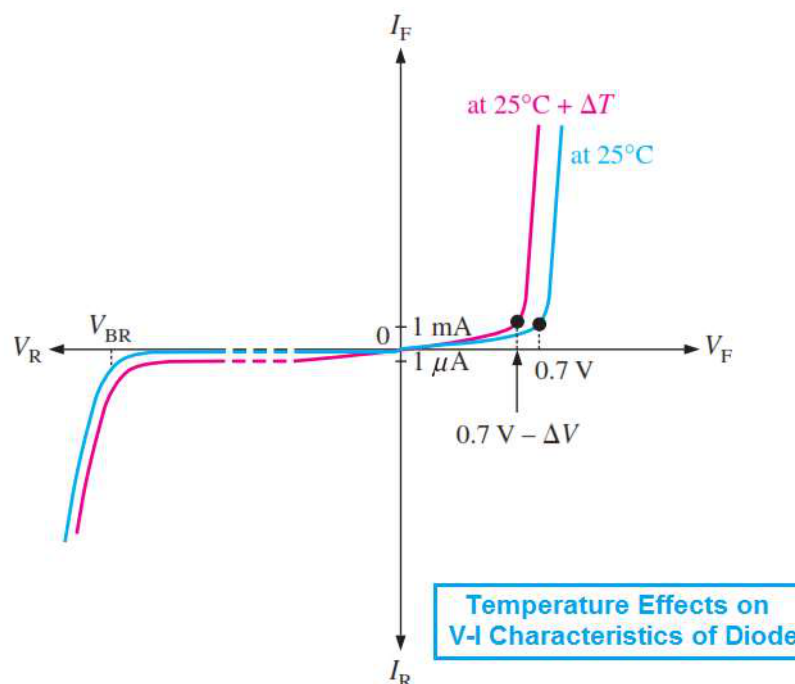


Figure 21: Effect of temperature on the diode characteristics — the forward curve shifts left as temperature rises.

Temperature dependence of reverse saturation current

The reverse saturation current approximately **doubles for every 10°C rise** in temperature:

$$I_s \text{ is approximately } I_{s0} \times 2^{(T - T_0)/10}$$

Effect on forward bias characteristics

- In the forward bias region, the silicon diode characteristic shifts to the left (lower voltage) at a rate of approximately 2.5 mV per °C increase in temperature.
- Forward bias voltage decreases with increasing temperature.

3.9 Breakdown Mechanisms

1. Avalanche Breakdown

- Occurs in lightly doped diodes where the depletion region is wide.
- A very large electric field at the junction ruptures covalent bonding, generating a large number of charge carriers — this results in **avalanche multiplication**.
- Temperature characteristic: **positive** — more voltage is required at higher temperature.

2. Zener Breakdown

- Occurs in heavily doped diodes where the depletion region is narrow.
- Even a small reverse voltage at the junction ruptures covalent bonding and generates a large number of charge carriers — this is the **Zener mechanism**.
- Temperature characteristic: **negative** — less voltage is required at higher temperature.

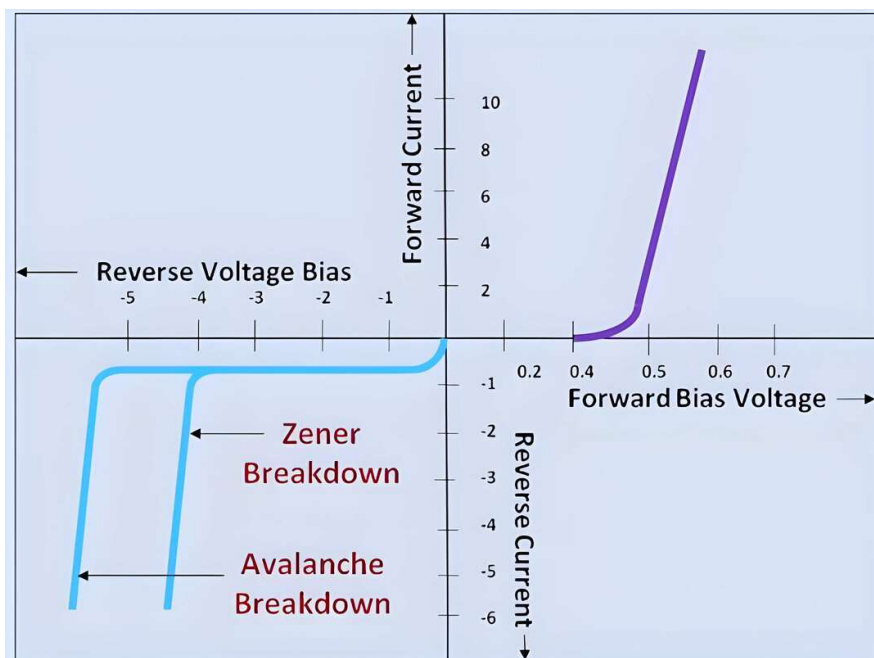


Figure 22: I–V characteristic of a Zener diode — note the sharp, nearly vertical breakdown region in reverse bias.

Property	Zener Breakdown	Avalanche Breakdown
Doping level	Heavily doped	Lightly doped
Depletion width	Narrow	Wide
Voltage required	Low (typically < 5 V)	High

Property	Zener Breakdown	Avalanche Breakdown
Temperature coefficient	Negative	Positive

3.10 Diode Capacitance

Depending upon the biasing condition, two types of capacitive effects exist in diodes:

A. Junction Capacitance (Depletion Capacitance), C_j

- The most significant capacitance; varies with the reverse bias voltage applied.
- Associated with the depletion region of the diode.
- As the reverse bias voltage increases, the depletion region widens, so the junction capacitance decreases.

$$C_j = \frac{\epsilon A}{W}$$

Figure 23: Junction capacitance formula.

$$C_j = (\text{permittivity of the dielectric} \times \text{junction area } A) / \text{depletion width } W$$

B. Diffusion Capacitance, C_d

- Appears in a forward-biased PN junction due to storage of charge in the diffusion region.
- As forward bias increases, the depletion region narrows and the concentration of injected charge carriers increases.
- This rate of change of injected charge with applied voltage is defined as diffusion capacitance.
- Increases with forward current; significant only when the diode is forward-biased.
- Typically less than junction capacitance in most applications.

$$C_d = \frac{\tau I_D}{V_T}$$

Figure 24: Diffusion capacitance formula.

$$C_d = (\text{carrier lifetime} \times I_D) / V_T \quad (I_D = \text{diode current, } V_T = \text{thermal voltage})$$

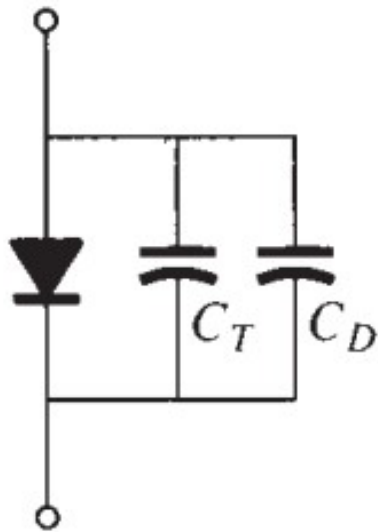


Figure 25: Equivalent circuit of a diode showing the transition (junction) and diffusion capacitances.

3.11 Equivalent Model of a Diode

The forward voltage V_F applied across the actual diode must overcome two quantities: (i) the potential barrier V_0 , and (ii) the internal voltage drop $I_f r_f$. Therefore:

$$V_F = V_0 + I_f r_f$$

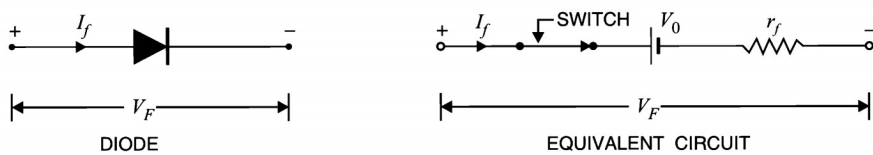


Figure 26: The diode forward voltage must overcome the barrier potential and the internal drop.

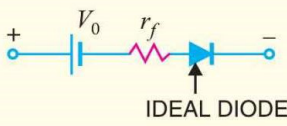
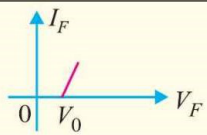
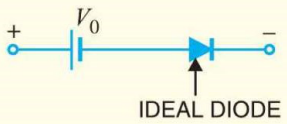
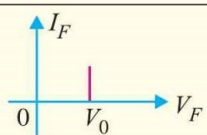
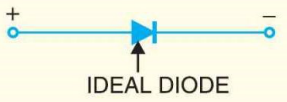
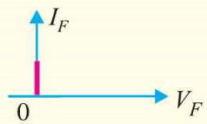
S.No.	Type	Model	Characteristic
1.	Approximate model	 IDEAL DIODE	
2.	Simplified model	 IDEAL DIODE	
3.	Ideal Model	 IDEAL DIODE	

Figure 27: The three diode models (approximate, simplified and ideal) with their I–V characteristics.

S.No.	Type	Characteristic / Description
1	Approximate Model	Includes ideal diode + barrier potential V_0 + internal dynamic resistance r_f . Forward voltage: $V_F = V_0 + I_f r_f$

S.No.	Type	Characteristic / Description
2	Simplified Model	Includes ideal diode + barrier potential V_0 only (r_f is neglected). Suitable for most analysis.
3	Ideal Model	Diode treated as a perfect switch: zero forward voltage drop, infinite reverse resistance. Used for quick estimates.

4. Zener Diode — Operation and Applications

4.1 What is a Zener Diode?

A **Zener diode** is a specially designed, heavily doped PN junction diode that is made to operate in the **reverse breakdown region** without getting damaged. While an ordinary diode is damaged by breakdown, a Zener diode is designed to work continuously in this region. Its circuit symbol is shown below.

Zener diode symbol



Figure 28: Circuit symbol of the Zener diode — the bent bar on the cathode distinguishes it from an ordinary diode.

4.2 Operation

The Zener diode is connected in **reverse bias** (its cathode to the positive side of the supply). Because the diode is heavily doped, the depletion region is very narrow, so Zener breakdown occurs at a low reverse voltage. Once the reverse voltage reaches the Zener voltage V_Z :

- The diode enters the breakdown region and conducts in reverse.
- The voltage across the Zener remains almost **constant at V_Z** even though the current through it changes — this is the key property.
- In the forward direction it behaves like an ordinary diode (0.7 V drop for Si).
- Zener breakdown has a negative temperature coefficient (see Section 3.9).

The full I–V characteristic of the Zener diode is shown in Section 3.9 — note the sharp, almost vertical breakdown region in reverse bias: this is where the Zener is designed to operate.

4.3 Zener Diode as a Voltage Regulator

The most common application of a Zener diode is to keep the output voltage of a power supply constant (regulated). The Zener is connected in parallel with the load, and a series resistor R_S limits the current.

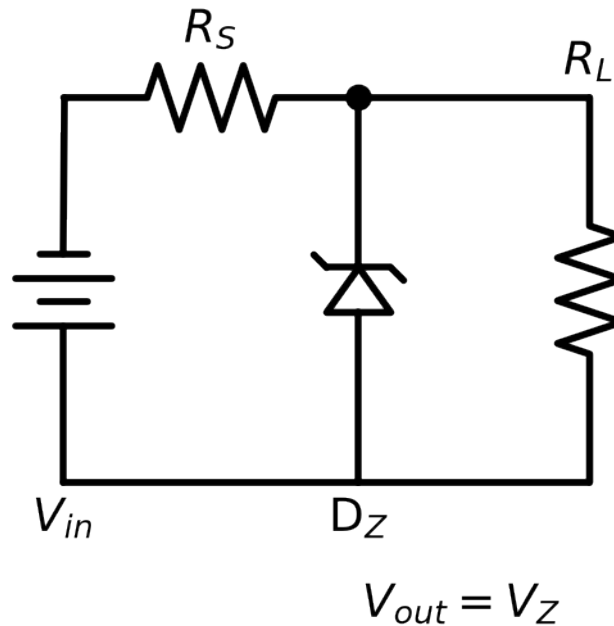


Figure 29: Zener voltage regulator — the Zener, in parallel with the load, holds the output at V_Z

Working (step by step):

- 1 The unregulated input voltage V_{in} (which must be greater than V_Z) is applied through the series resistor R_S .
- 2 The Zener is in reverse bias and breaks down, so the voltage across it — and hence across the load — stays at V_Z . The output is therefore $V_{out} = V_Z$.
- 3 If the input voltage increases, the extra voltage appears across R_S (the Zener draws more current), and the output still stays at V_Z .
- 4 If the load draws more current, the Zener automatically passes less current — the total current through R_S adjusts itself while the output remains V_Z .
- 5 R_S must be chosen so that the Zener current stays within its safe limits at maximum input voltage and minimum load current.

4.4 Applications of the Zener Diode

- **Voltage regulation** — keeping a DC output voltage constant.
- **Overvoltage protection** — protecting sensitive circuits from voltage spikes.
- **Reference voltage source** — a fixed reference for comparators and meters.
- **Wave shaping / clipping circuits** — clipping signals at precise levels.
- **Meter protection** — protecting measurement instruments.

5. Applications of the PN Junction & Diode as a Rectifier

5.1 Applications Overview

Application	Function
Diodes	Allow current to flow in only one direction — signal processing, protection
Rectifiers	Convert AC to DC in power supply circuits
LEDs	When forward biased, energy is released as light — displays, indicators
Solar Cells	Convert light energy into electrical energy using the photovoltaic effect
Transistors	Form the basis of BJTs and FETs

5.2 Rectifier Circuits

A **rectifier** converts an AC signal into a DC signal. A complete power supply system includes four stages:

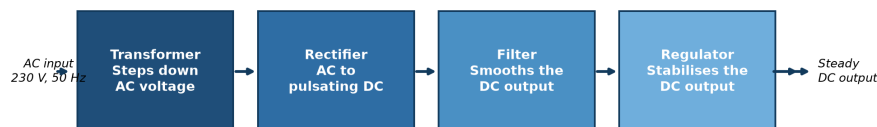


Figure 30: Block diagram of a complete DC power supply.

- **Transformer** — steps down the high voltage AC from the mains to a lower, usable voltage level (e.g., 230 V AC to 12 V AC).
- **Rectifier** — converts AC to a unipolar (one-direction) output using diodes.
- **Filter** — converts the pulsating rectified output to a nearly constant DC using capacitors.
- **Regulator** — reduces the ripple and stabilises the DC output voltage.

5.3 Half-Wave Rectifier

A half-wave rectifier uses a **single diode**. Only the positive half-cycle of the AC passes through to the load; the negative half is blocked.

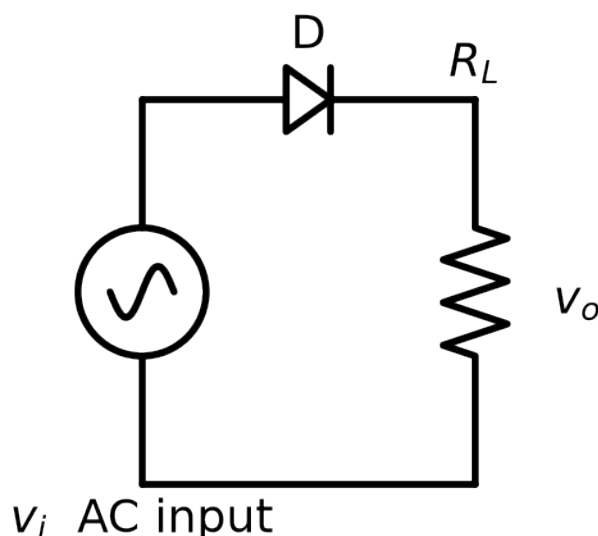


Figure 31: Half-wave rectifier circuit.

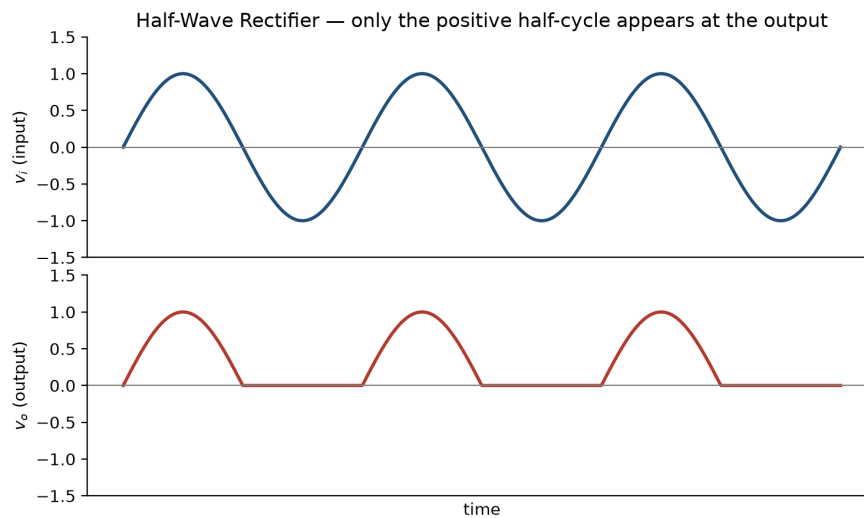


Figure 32: Half-wave rectifier — input and output waveforms (the negative half is blocked).

Key components and their functions

- **Transformer** — steps down the AC voltage to a usable level.
- **Signal diode (rectifier)** — acts as a one-way valve. It allows current only when the anode is positive relative to the cathode (forward bias), and clips off the negative half of the AC cycle.
- **Filter capacitor** — stores energy during peaks and releases it during troughs, smoothing the pulsating DC into a nearly constant voltage.
- **Load** — the component that consumes DC power. The output voltage is measured across the load.

Half-wave rectifier — performance

Parameter	Value
Ripple factor	1.21 (high ripple — poor quality DC)
Maximum efficiency	40.6%
Output frequency	Same as input frequency (f_{in})
PIV rating of the diode	V_m (peak input voltage)

5.4 Full-Wave Rectifier

A full-wave rectifier utilises **both half-cycles** of the AC waveform. Two main configurations exist: the centre-tapped full-wave rectifier and the bridge full-wave rectifier.

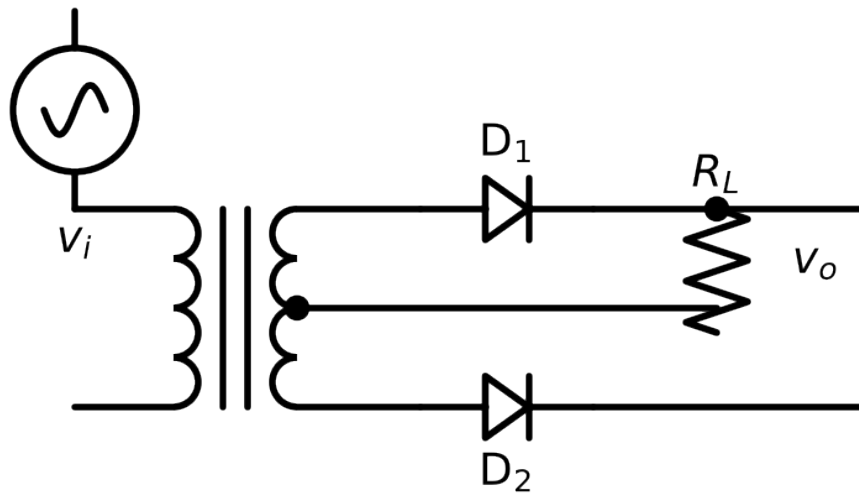


Figure 33: Centre-tapped full-wave rectifier — D_1 conducts on one half-cycle, D_2 on the other.

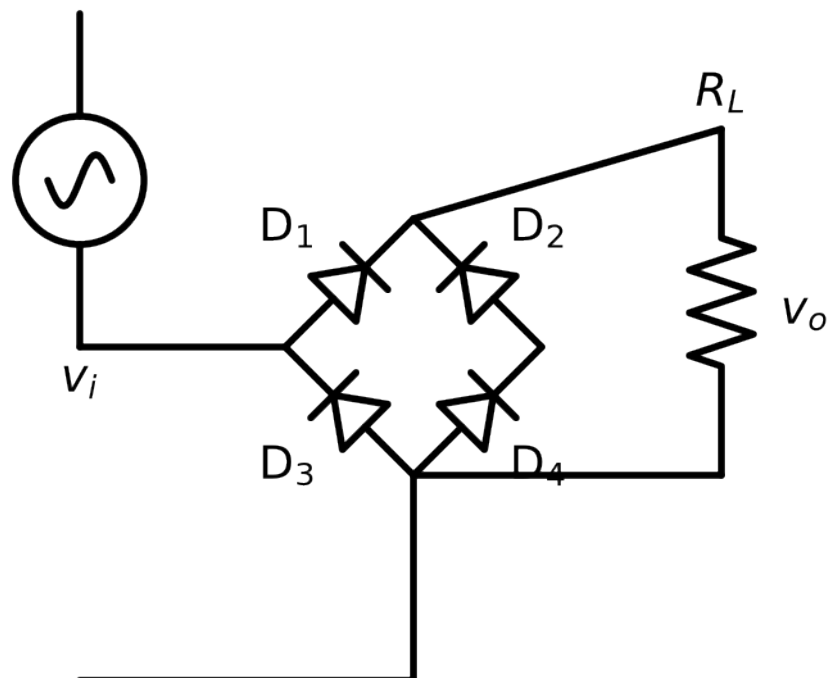


Figure 34: Bridge full-wave rectifier — four diodes; a standard transformer is enough.

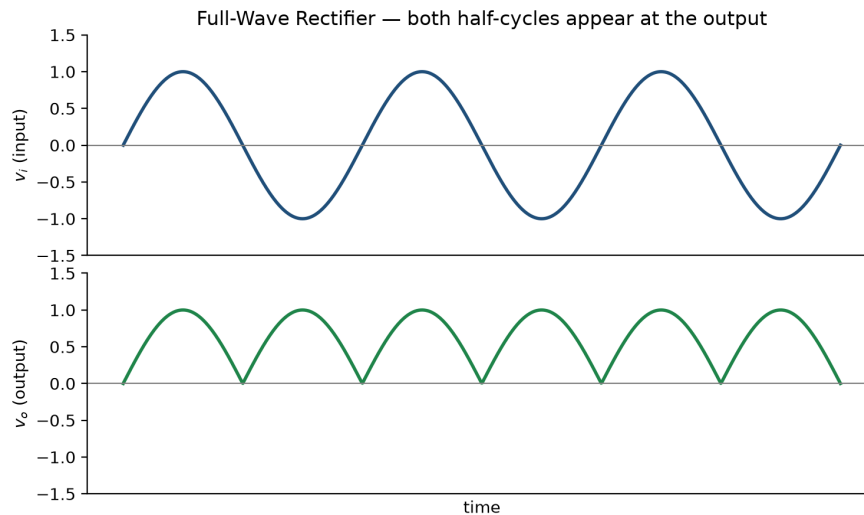


Figure 35: Full-wave rectifier — both half-cycles appear at the output (negative half is flipped up).

Feature	Centre-Tapped Full-Wave	Bridge Full-Wave
Number of diodes	2	4
Transformer	Requires centre-tapped (bulkier & expensive)	Standard (smaller & cheaper)
Peak Inverse Voltage (PIV)	$2V_m$	V_m
Transformer Utilisation (TUF)	About 0.693 (low)	About 0.812 (higher)
Output frequency	$2 \times f_{in}$	$2 \times f_{in}$
Maximum efficiency	81.2%	81.2%
Cost	Higher (specialised transformer)	Lower (standard components)
Suitability	Low voltage, high current	General-purpose DC power supplies

5.5 Rectifier Performance Metrics

1. Ripple Factor

The ripple factor measures the amount of AC component (unwanted variation) remaining in the DC output. A lower value is better.

$$\text{Ripple factor} = [(V_{rms} / V_{dc})^2 - 1]^{1/2}$$

- Half-wave: ripple factor = 1.21
- Full-wave: ripple factor = 0.48

2. Rectifier Efficiency

The ratio of DC output power to applied AC input power:

$$\text{Efficiency} = (P_{dc} / P_{ac}) \times 100\%$$

- Half-wave: maximum 40.6%
- Full-wave: maximum 81.2%

3. Peak Inverse Voltage (PIV)

The maximum reverse voltage a diode must withstand without breaking down:

- Half-wave: V_m

- Full-wave (centre-tap): $2V_m$
- Full-wave (bridge): V_m

5.6 Summary: Half-Wave vs. Full-Wave Rectifier

Parameter	Half-Wave Rectifier	Full-Wave Rectifier (Bridge & Centre-Tap)
Ripple factor	1.21 (high ripple, poor DC quality)	0.48 (low ripple, smoother DC)
Maximum efficiency	40.6%	81.2%
Output frequency	f_{in} (same as input, e.g., 50 Hz)	$2 \times f_{in}$ (double, e.g., 100 Hz)
PIV rating	V_m (peak voltage)	Bridge: V_m Centre-tap: $2V_m$
Number of diodes	1	Bridge: 4 Centre-tap: 2
Transformer type	Standard	Bridge: standard Centre-tap: special (bulkier)

6. Diode Clippers and Clampers

6.1 Diode Clippers (Limiters)

A **clipper** is a circuit that removes (clips) a portion of the input signal — positive half, negative half, or both — without distorting the remaining part. It is used for protection or waveform shaping.

Type 1: Series Clipper

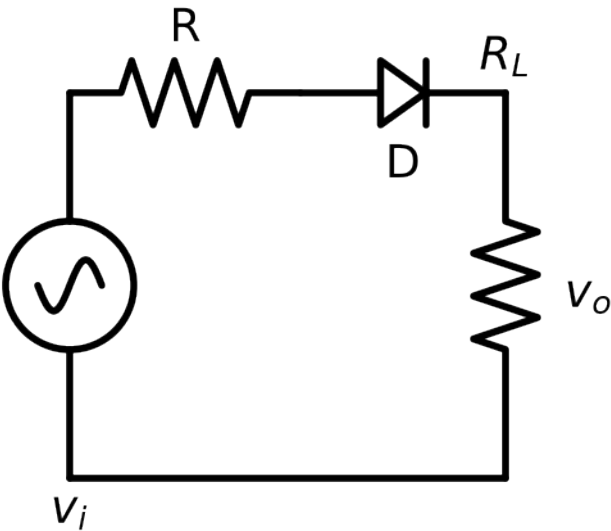


Figure 36: Series clipper — the diode is in series with the load resistor.

- The diode is placed in series with the load resistor.
- When the diode is OFF: no current flows, the output is zero (clipped).
- When the diode is ON: the input signal passes through to the output.
- Reversing the diode clips the opposite half-cycle of the output.

Type 2: Shunt (Parallel) Clipper

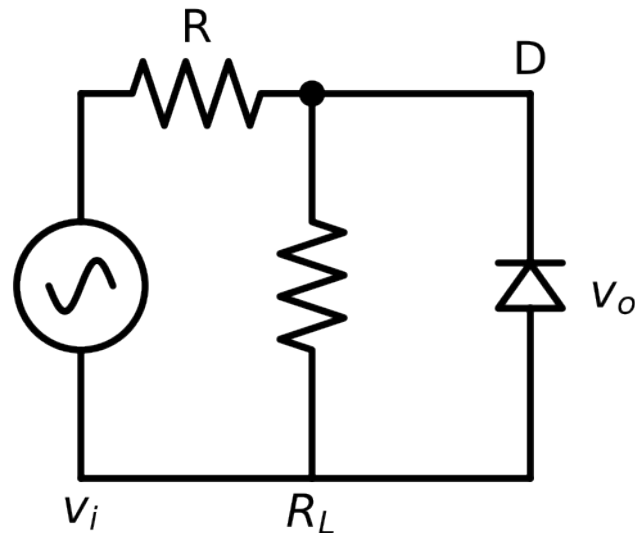


Figure 37: Shunt (parallel) clipper — the diode is in parallel with the load.

- The diode is placed in parallel with the load.
- When the diode is ON: it short-circuits the signal (clips it).
- When the diode is OFF: the signal passes through to the load.

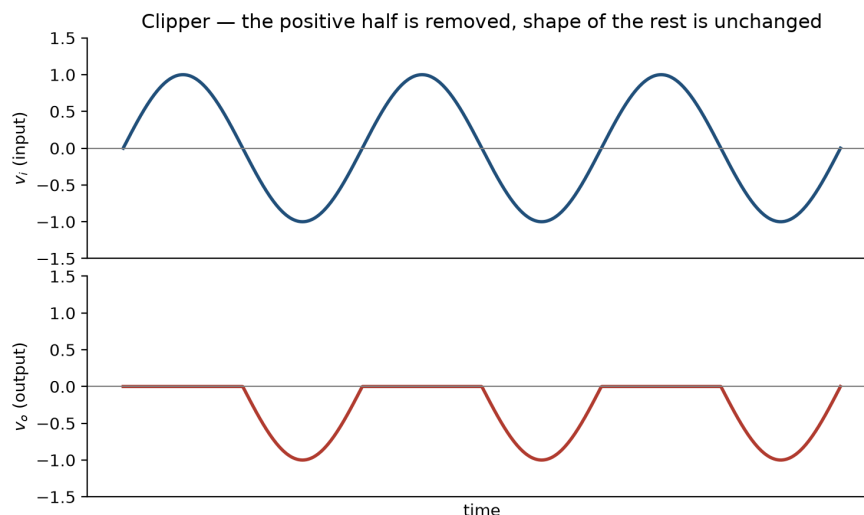


Figure 38: Clipping action — the positive half is removed; the remaining waveform keeps its shape.

6.2 Diode Clamper (DC Restorers)

A **clamper** is a circuit that shifts the entire waveform up or down by adding a DC component to the signal. Unlike a clipper, it does **not** change the shape of the waveform — it only changes the DC level (baseline).

- **Key component required:** a capacitor is essential for clamper operation.

Type 1: Positive Clamper

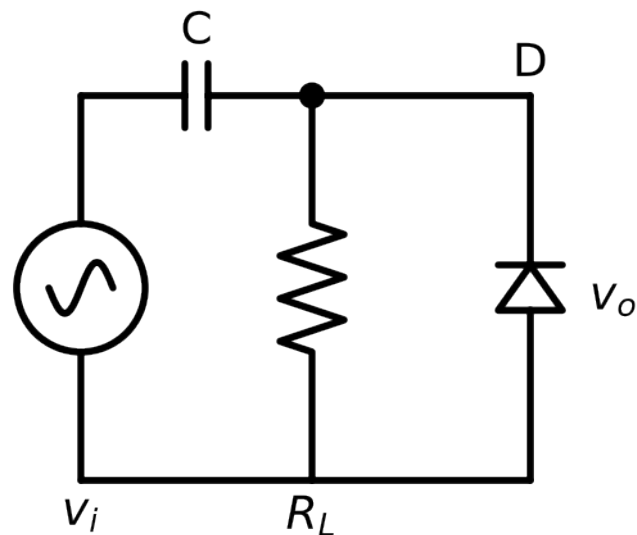


Figure 39: Positive clamper circuit — capacitor in series, diode in parallel with the load.

- Shifts the waveform upward.
- The negative peak of the signal is clamped to zero (or a reference voltage).
- The entire waveform sits above the x-axis.

Type 2: Negative Clamper

- Shifts the waveform downward.
- The positive peak is clamped to zero.
- The entire waveform sits below the x-axis.

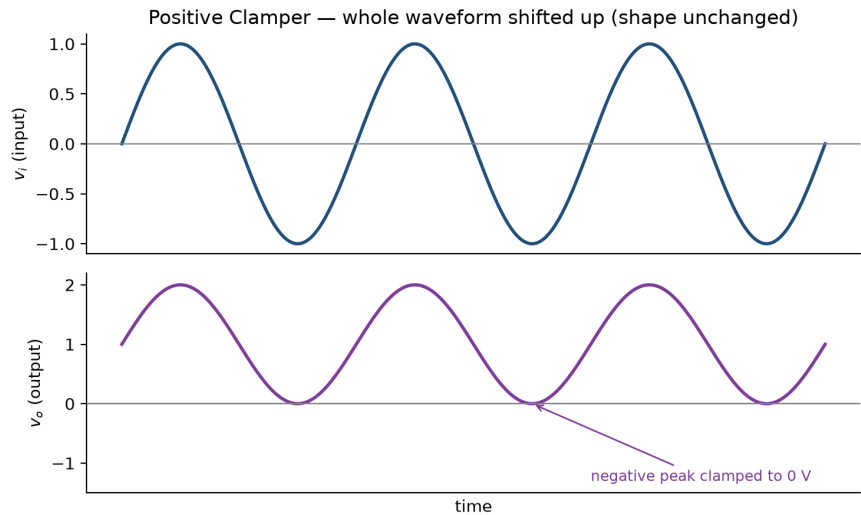


Figure 40: Positive clamper — the waveform is shifted up; its shape is unchanged.

6.3 Clipper vs. Clamper

Feature	Clipper	Clamper
Function	Removes part of the waveform	Shifts the waveform up or down
Waveform shape	Changed (distorted)	Unchanged (shape preserved)
Key component	Diode (+ resistor)	Diode + capacitor (essential)
Output level	Clipped at threshold	DC shifted

7. Solved Numerical Examples

Work through these examples step by step — they cover every type of numerical problem asked from this module.

Example 1: Thermal Voltage Calculation

Problem: At a temperature of 27°C, determine the thermal voltage V_T . Given $k = 1.38 \times 10^{-23}$ J/K and $q = 1.6 \times 10^{-19}$ C.

Solution:

$$T_K = 273 + 27 = 300 \text{ K}$$

$$V_T = (k \times T_K) / q = (1.38 \times 10^{-23} \times 300) / (1.6 \times 10^{-19}) = 25.875 \text{ mV, i.e. about 26 mV}$$

$$T = 273 + ^\circ\text{C} = 273 + 27 = 300 \text{ K}$$

$$V_T = \frac{kT_K}{q} = \frac{(1.38 \times 10^{-23} \text{ J/K})(300 \text{ K})}{1.6 \times 10^{-19} \text{ C}}$$

$$= 25.875 \text{ mV} \cong 26 \text{ mV}$$

Figure 41: Thermal voltage calculation at room temperature.

Example 2: Series Diode Circuit — Forward Biased

Problem: For a series diode configuration with $E = 8 \text{ V}$ and $R = 2.2 \text{ k}\Omega$ (silicon diode), determine V_D , V_R and I_D .

Solution:

The applied voltage is sufficient to turn the silicon diode ON.

$$V_D = 0.7 \text{ V (barrier potential of silicon)}$$

$$V_R = E - V_D = 8 - 0.7 = 7.3 \text{ V}$$

$$I_D = V_R / R = 7.3 \text{ V} / 2200 \Omega = 3.32 \text{ mA}$$

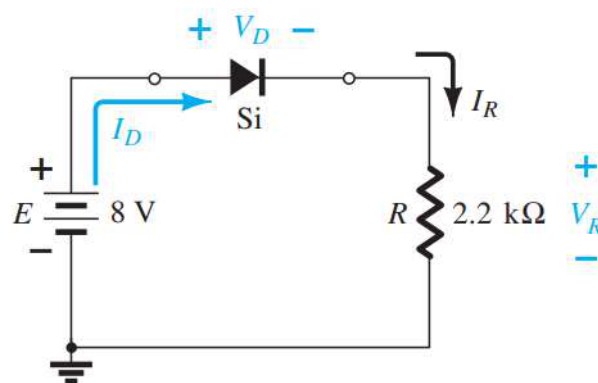


Figure 42: The given series diode circuit.

Solution: Since the applied voltage establishes a current in the clockwise direction to match the arrow of the symbol and the diode is in the “on” state,

$$V_D = 0.7 \text{ V}$$

$$V_R = E - V_D = 8 \text{ V} - 0.7 \text{ V} = 7.3 \text{ V}$$

$$I_D = I_R = \frac{V_R}{R} = \frac{7.3 \text{ V}}{2.2 \text{ k}\Omega} \cong 3.32 \text{ mA}$$

Figure 43: Step-by-step solution.

Example 3: Series Diode Circuit — Insufficient Voltage

Problem: For a series silicon diode with $E = 0.5 \text{ V}$ and $R = 1.2 \text{ k}\Omega$, determine V_D , V_R and I_D .

Solution:

The applied voltage (0.5 V) is insufficient to turn the silicon diode ON (it requires 0.7 V).

$$I_D = 0 \text{ A}$$

$$V_R = I_D \times R = 0 \text{ A} \times 1200 \Omega = 0 \text{ V}$$

$$V_D = E = 0.5 \text{ V (all the voltage appears across the diode)}$$

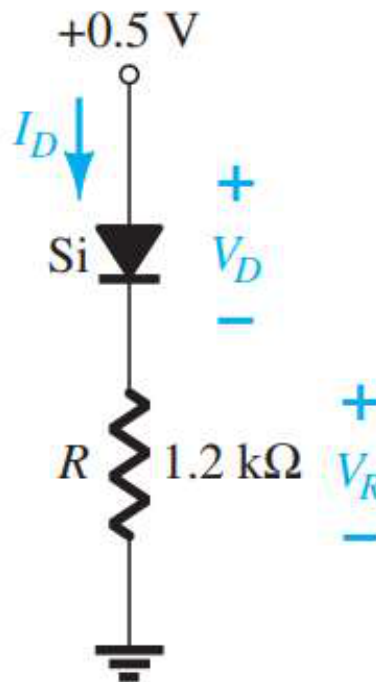


Figure 44: The given circuit — note the very small applied voltage.

$$I_D = 0 \text{ A}$$

$$V_R = I_R R = I_D R = (0 \text{ A}) 1.2 \text{ k}\Omega = 0 \text{ V}$$

$$V_D = E = 0.5 \text{ V}$$

Figure 45: Resulting voltage and current levels.

Example 4: Dual Diode Circuit — Finding the Current

Problem: In a circuit with $E_1 = 24 \text{ V}$, $E_2 = 4 \text{ V}$, $R = 2 \text{ k}\Omega$, D_1 (forward biased, Si) and D_2 (reverse biased), find the current I . Assume the forward resistance of the diodes is zero.

Solution:

D_1 is ON; D_2 is OFF (it acts as an open circuit).

$$I = (E_1 - E_2 - V_0) / R = (24 - 4 - 0.7) / 2000 = 19.3 / 2000 = 9.65 \text{ mA}$$

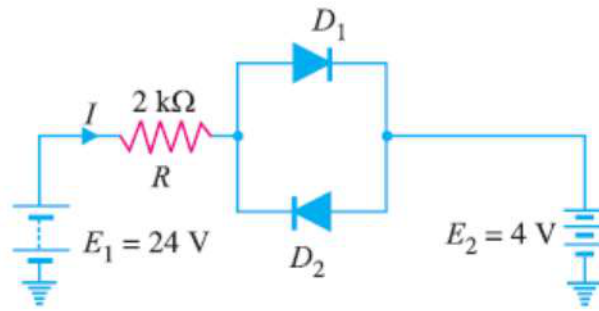
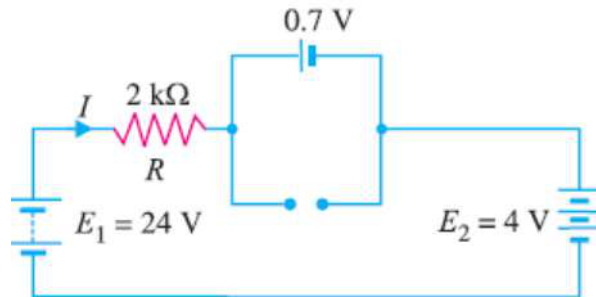


Figure 46: The given dual-diode circuit.

Figure 47: D_2 reverse biased $\rightarrow$ replaced by an open circuit.

$$I = \frac{E_1 - E_2 - V_0}{R} = \frac{24 - 4 - 0.7}{2 \text{ k}\Omega} = \frac{19.3 \text{ V}}{2 \text{ k}\Omega} = 9.65 \text{ mA}$$

Figure 48: Current calculation.

Example 5: Ideal Diode Circuit

Problem: A circuit has ideal diodes D_1 (reverse biased) and D_2 (forward biased), with a 10 V supply, $R_1 = 2 \text{ }\Omega$, $R_2 = 3 \text{ }\Omega$ and $R_3 = 2 \text{ }\Omega$. Find the current through R_1 .

Solution:

D_1 is reverse biased (open circuit); D_2 is forward biased (short circuit for an ideal diode).

$$R_{\text{eq}} = R_1 + R_3 = 2 + 2 = 4 \text{ }\Omega$$

$$\text{Current through } R_1 = V / R_{\text{eq}} = 10 / 4 = 2.5 \text{ A}$$

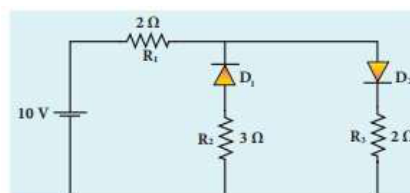
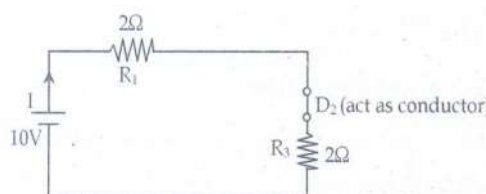


Figure 49: The given circuit with two ideal diodes.

Figure 50: Reduced circuit after replacing D_1 by an open circuit and D_2 by a short circuit.

$$\text{Effective resistance } R_{\text{eff}} = R_1 + R_3 = 4\Omega$$

$$\text{Current through } R_1 = V/R_{\text{eff}} = 10/4 = 2.5\text{A}$$

Figure 51: Effective resistance and current.

Example 6: Currents through the 3 Ω and 4 Ω Resistors

Problem: Determine the current flowing through the 3 Ω and 4 Ω resistors of the circuit shown below. Assume that the diodes D_1 and D_2 are ideal diodes.

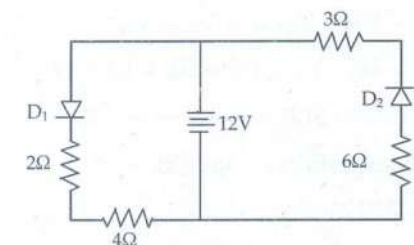
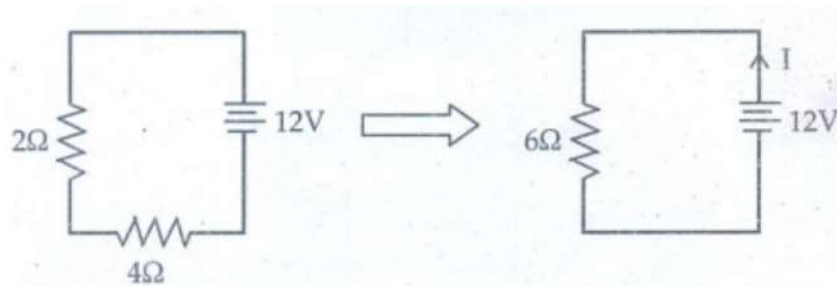
Solution:

From the figure, D_2 is reverse biased. So it does not conduct current.

Current through the 3 Ω resistor = 0

D_1 is forward biased, and it is an ideal diode (acts as a short circuit).

$$\text{Current through the 4 Ω resistor: } I = V/R = 12\text{ V} / 6\Omega = 2\text{ A}$$

Figure 52: The given circuit (12 V supply, diodes D_1 , D_2).Figure 53: The circuit after replacing D_1 (ideal, ON) by a short circuit.

The current through 4 Ω is

$$\therefore I = V/R = 12/6 = 2\text{A}$$

Figure 54: Current through the 4 ohm resistor.

Example 7: Voltage V_A with Ge and Si Diodes in Parallel

Problem: Find the voltage V_A in the circuit shown below (a germanium and a silicon diode in parallel). Use the simplified model.

Solution:

When the voltage is applied, the germanium diode ($V_0 = 0.3\text{ V}$) turns on first, and a level of 0.3 V is maintained across the parallel circuit.

The silicon diode never gets the opportunity to have 0.7 V across it and therefore remains in the open-circuit state.

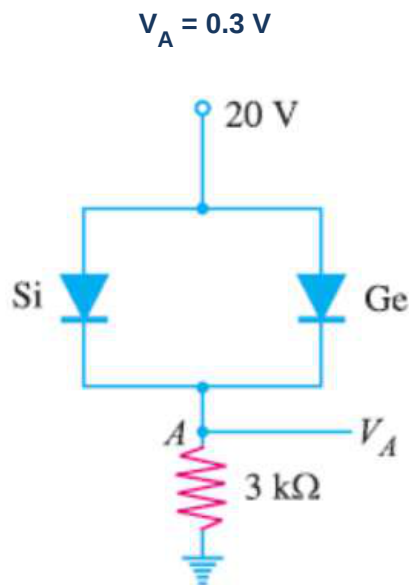


Figure 55: The given circuit — Si and Ge diodes in parallel with the output at node A.

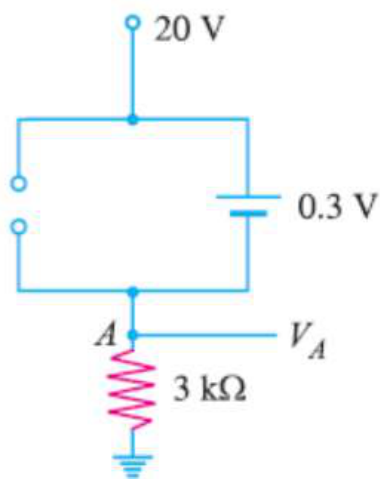


Figure 56: The Ge diode (0.3 V) clamps the node; the Si diode stays OFF.

Example 8: Determining the State of a Diode

Problem: Determine the state of the diode for the circuit shown below, and find I_D and V_D . Assume the simplified model for the diode.

Solution:

Assume the diode is ON. Replace it with a 0.7 V battery as shown in the figure.

$$I_1 = (2 - 0.7) \text{ V} / 4 \text{ k}\Omega = 1.3 \text{ V} / 4 \text{ k}\Omega = 0.325 \text{ mA}$$

$$I_2 = 0.7 \text{ V} / 1 \text{ k}\Omega = 0.7 \text{ mA}$$

$$I_D = I_1 - I_2 = 0.325 - 0.7 = -0.375 \text{ mA}$$

Since the diode current is negative, the diode must be OFF, and the true value of the diode current is $I_D = 0 \text{ mA}$. The initial assumption was wrong.

To analyse the circuit properly, replace the diode with an open circuit. The voltage V_D across the diode is:

$$V_D = 2 \text{ V} \times 1 \text{ k}\Omega / (1 \text{ k}\Omega + 4 \text{ k}\Omega) = 0.4 \text{ V}$$

We know that 0.7 V is required to turn the diode ON. Since V_D is only 0.4 V, the answer confirms that the diode is OFF.

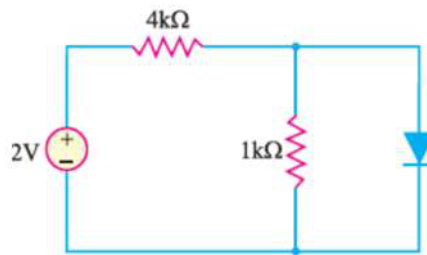


Figure 57: The given circuit.

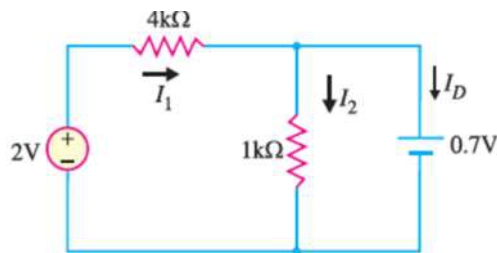


Figure 58: Assuming the diode is ON → replaced by a 0.7 V battery.

$$I_1 = \frac{(2 - 0.7) \text{ V}}{4 \text{ k}\Omega} = \frac{1.3 \text{ V}}{4 \text{ k}\Omega} = 0.325 \text{ mA}$$

$$I_2 = \frac{0.7 \text{ V}}{1 \text{ k}\Omega} = 0.7 \text{ mA}$$

$$I_D = I_1 - I_2 = 0.325 - 0.7 = -0.375 \text{ mA}$$

Figure 59: Calculating I_1 , I_2 and I_D .

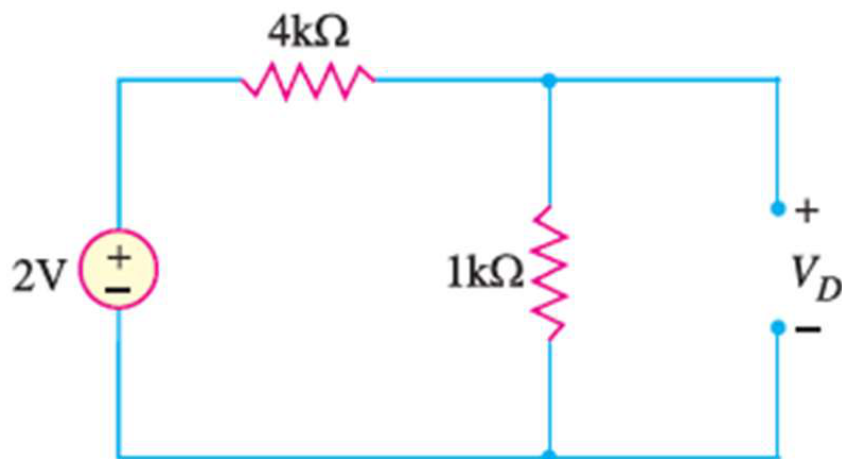


Figure 60: Analysing properly: the diode is replaced by an open circuit.

$$V_D = \frac{2 \text{ V}}{1 \text{ k}\Omega + 4 \text{ k}\Omega} \times 1 \text{ k}\Omega = 0.4 \text{ V}$$

Figure 61: The diode voltage confirms it is OFF.

Example 9: Shockley Equation — Germanium Diode

Problem: A germanium diode has a reverse saturation current $I_0 = 0.19 \mu\text{A}$. Find the current when it is forward biased with 0.3 V at 27°C. ($n = 1$ for Ge.)

Solution:

Step 1 — thermal voltage: $V_T = T_K / 11600 = (273 + 27) / 11600 = 300 / 11600 = 25.862 \text{ mV}$

Step 2 — apply Shockley's equation:

$$I_D = 0.19 \times 10^{-6} \times (e^{0.3 / (1 \times 25.862 \text{ mV})} - 1) = 20.72 \text{ mA}$$

Given:

$I_0 = 0.19 \mu\text{A}$, $V_D = 0.3\text{V}$, $T = 27^\circ\text{C}$, $\eta = 1$ for Germanium Diode, and $I_D = ?$

$$I_D = I_0 \left(e^{V_D / \eta V_T} - 1 \right)$$

$$\Rightarrow I_D = 0.19 \times 10^{-6} \times \left(e^{0.3\text{V} / 1 \times 25.862 \times 10^{-3}} - 1 \right)$$

$$\Rightarrow I_D = 20.72 \text{ mA}$$

$$V_T = \frac{T}{11,600}$$

$$= \frac{273^\circ + 27^\circ}{11,600}$$

$$= 25.862 \text{ mV}$$

Figure 62: Forward bias current calculation for the germanium diode.

Example 10: Germanium Diode — Reverse Saturation Current and Bias Voltage

Problem: A germanium diode carries a current of 10 mA when it is forward biased with 0.2 V at 27°C.

(a) Find the reverse saturation current. (b) Find the bias voltage required to get a current of 100 mA.

Solution:

Given: $I_D = 10 \text{ mA}$, $V_D = 0.2 \text{ V}$, $T = 27^\circ\text{C}$, $n = 1$ (Ge).

$$V_T = T_K / 11600 = 300 / 11600 = 25.862 \text{ mV}$$

$$(a) I_0 = I_D / (e^{V_D / \eta V_T} - 1) = 10 \text{ mA} / (e^{0.2 / (1 \times 25.862 \text{ m})} - 1) = 4.3816 \mu\text{A}$$

$$(b) V_D = \eta V_T \ln(I_D / I_0 + 1) = 1 \times 25.862 \text{ mV} \times \ln(100 \text{ mA} / 4.3816 \mu\text{A} + 1) = 0.2595 \text{ V}$$

Given, $I_D = 10 \text{ mA}$, $V_D = 0.2 \text{ V}$, $t = 27^\circ\text{C}$, $\eta = 1$ for Ge Diode

(a) $I_0 = ?$

$$I_0 = \frac{I_D}{\left(e^{V_D / \eta V_T} - 1 \right)}$$

$$I_0 = \frac{10 \text{ m}}{\left(e^{0.2 / 1 \times 25.862 \text{ m}} - 1 \right)} \Rightarrow I_0 = 4.3816 \mu\text{A}$$

$$V_T = \frac{T}{11,600}$$

$$= \frac{273^\circ + 27^\circ}{11,600}$$

$$= 25.862 \text{ mV}$$

Figure 63: Part (a): finding the reverse saturation current.

(b) $I_D = 100 \text{ mA}$, $V_D = ?$

$$\Rightarrow V_D = \eta V_T \ln \left(\frac{I_D}{I_0} + 1 \right)$$

$$\Rightarrow V_D = 1 \times 25.862 \text{ mV} \times \ln \left(\frac{100 \text{ m}}{4.3816 \mu} + 1 \right)$$

$$V_D = 0.2595 \text{ V}$$

Figure 64: Part (b): finding the bias voltage for 100 mA.

Example 11: Forward Bias Voltage — Silicon Diode

Problem: The forward current in a silicon diode is 15 mA at 27°C. If the reverse saturation current is 0.24 nA, find the forward bias voltage. ($n = 2$ for Si.)

Solution:

Given: $I_D = 15 \text{ mA}$, $I_0 = 0.24 \text{ nA}$, $T = 27^\circ\text{C}$, $n = 2$.

$$V_T = T_K / 11600 = 300 / 11600 = 25.862 \text{ mV}$$

$$V_D = n V_T \ln(I_D / I_0 + 1) = 2 \times 25.862 \times 10^{-3} \times \ln(15 \text{ mA} / 0.24 \text{ nA} + 1) = 0.928 \text{ V}$$

$$I_D = 15 \text{ mA}, I_0 = 0.24 \text{ nA}, T = 27^\circ\text{C}, \eta = 2 \text{ for Si Diode, and } V_D = ?$$

$$I_D = I_0 \left(e^{V_D / \eta V_T} - 1 \right)$$

$$\Rightarrow V_D = \eta V_T \ln \left(\frac{I_D}{I_0} + 1 \right)$$

$$\Rightarrow V_D = 2 \times 25.862 \text{ mV} \times \ln \left(\frac{15 \text{ m}}{0.24 \text{ n}} + 1 \right)$$

$$V_D = 0.928 \text{ V}$$

$$V_T = \frac{T}{11,600}$$

$$= \frac{273^\circ + 27^\circ}{11,600}$$

$$= 25.862 \text{ mV}$$

Figure 65: Forward bias voltage calculation for the silicon diode.

Example 12: Silicon Diode — Saturation Current, Voltage and Temperature

Problem: A silicon diode has a saturation current of 1 pA at 20°C. Determine (a) the diode bias voltage when the diode current is 3 mA, and (b) the diode bias current when the temperature changes to 100°C, for the same bias voltage.

Solution:

Given: $I_0 = 1 \text{ pA}$, $I_D = 3 \text{ mA}$, $t = 20^\circ\text{C}$, $n = 2$ (Si).

$$V_T = T_K / 11600 = (273 + 20) / 11600 = 25.2586 \text{ mV}$$

$$(a) V_D = n V_T \ln(I_D / I_0 + 1) = 2 \times 25.2586 \text{ mV} \times \ln(3 \text{ mA} / 1 \text{ pA} + 1) = 1.102 \text{ V}$$

(b) At 100°C: the saturation current doubles every 10°C:

$$I_{02} = I_{01} \times 2^{(t_2 - t_1) / 10} = 10^{-12} \times 2^{(100 - 20) / 10} = 256 \text{ pA}$$

$$V_T \text{ at } 100^\circ\text{C} = (273 + 100) / 11600 = 32.155 \text{ mV}$$

$$I_D = 256 \text{ pA} \times (e^{1.102 / (2 \times 32.155 \text{ m})} - 1) = 7.08 \text{ mA}$$

Given, $I_0 = 1 \text{ pA}$, $I_D = 3 \text{ mA}$, $t = 20^\circ\text{C}$, $\eta = 2$ for Si Diode.

$$(a) V_D = ? \quad V_D = \eta V_T \ln \left(\frac{I_D}{I_0} + 1 \right)$$

$$\Rightarrow V_D = 2 \times 25.2586 \text{ mV} \times \ln \left(\frac{3 \text{ m}}{1 \text{ p}} + 1 \right)$$

$$V_D = 1.102 \text{ V}$$

$$V_T = \frac{T}{11,600}$$

$$= \frac{273^\circ + 20^\circ}{11,600}$$

$$= 25.2586 \text{ mV}$$

Figure 66: Part (a): bias voltage at 3 mA.

$$(b) \ t = 100^{\circ}\text{C} \quad I_D = ?$$

$$I_{02} = I_{01} \cdot 2^{(t_2 - t_1)/10} = 10^{-12} \times 2^{(100 - 20)/10} = 256 \text{ pA}$$

$$\Rightarrow I_D = 256 \text{ pA} \times \left(e^{1.102 / (2 \times 32.155 \text{ m})} - 1 \right)$$

$$\Rightarrow I_D = 7.08 \text{ mA}$$

$$V_T = \frac{T}{11,600} = \frac{273^{\circ} + 100^{\circ}}{11,600} = 32.155 \text{ mV}$$

Figure 67: Part (b): current at 100°C for the same bias voltage.

Example 13: Static and Dynamic Resistance — Germanium Diode

Problem: Calculate the static and dynamic resistance of a PN junction diode when $V_D = 0.25 \text{ V}$ for a germanium diode. $I_0 = 1 \text{ } \mu\text{A}$, $T = 300 \text{ K}$, $n = 1$.

Solution:

$$V_T = T_K / 11600 = 300 / 11600 = 25.862 \text{ mV}$$

$$I_D = 1 \text{ } \mu\text{A} \times (e^{0.25 / (1 \times 25.862 \text{ mV})} - 1) = 15.782 \text{ mA}$$

$$(a) \text{ Static resistance: } R_D = V_D / I_D = 0.25 \text{ V} / 15.782 \text{ mA} = 15.84 \text{ } \blacksquare$$

$$(b) \text{ Dynamic resistance: } r_d = n V_T / I_D = 1 \times 25.862 \text{ mV} / 15.782 \text{ mA} = 1.638 \text{ } \blacksquare$$

Given, $I_0 = 1 \text{ } \mu\text{A}$, $V_D = 0.25 \text{ V}$, $T = 300 \text{ K}$, $\eta = 1$ for Ge Diode

$$I_D = I_0 \left(e^{V_D / \eta V_T} - 1 \right)$$

$$\Rightarrow I_D = 1 \text{ } \mu\text{A} \times \left(e^{0.25 / (1 \times 25.862 \text{ m})} - 1 \right)$$

$$\Rightarrow I_D = 15.782 \text{ mA}$$

$$V_T = \frac{T}{11,600} = \frac{273^{\circ} + 27^{\circ}}{11,600} = 25.862 \text{ mV}$$

Figure 68: Finding the diode current.

For Static Resistance,

$$(a) \ R_D = \frac{V_D}{I_D} = \frac{0.25}{15.782 \text{ m}} = 15.84 \text{ } \Omega$$

For Dynamic Resistance,

$$(b) \ r_d = \frac{\eta V_T}{I_D} = \frac{1 \times 25.862 \text{ m}}{15.782 \text{ m}} = 1.638 \text{ } \Omega$$

Figure 69: Calculating the static and dynamic resistance.

8. Quick Reference — Key Formulae and Definitions

8.1 Key Formulae Summary

Quantity	Formula	Notes
Diode current (Shockley)	$I_D = I_s (e^{V_D/nV_T} - 1)$	$n = 1$ (Ge), $n = 2$ (Si)
Thermal voltage	$V_T = k T_K / q$	About 26 mV at 27°C
DC (static) resistance	$R_D = V_D / I_D$	At a specific operating point
AC (dynamic) resistance	$r_d = n V_T / I_D = 26 \text{ mV} / I_D$ ($n = 1$)	Slope of the I–V curve
Junction capacitance	$C_j = (\text{permittivity} \times A) / W$	Reverse biased
Diffusion capacitance	$C_d = (\text{carrier lifetime} \times I_D) / V_T$	Forward biased
Ripple factor	$[(V_{rms}/V_{dc})^2 - 1]^{1/2}$	HW: 1.21, FW: 0.48
Rectifier efficiency	$(P_{dc} / P_{ac}) \times 100\%$	HW: 40.6%, FW: 81.2%
Reverse current vs. temperature	$I_s = I_{s0} \times 2^{(T-T_0)/10}$	Doubles every 10°C
Diode forward voltage (model)	$V_F = V_0 + I_f r_f$	Approximate model

8.2 Key Definitions at a Glance

Term	Definition
Semiconductor	A material with electrical conductivity between conductors and insulators; conductivity can be controlled by doping or external fields.
Doping	The deliberate addition of impurities to a pure semiconductor to alter its electrical properties.
Depletion region	A region at the PN junction devoid of free charge carriers, containing only immobile ions.
Barrier potential (V_b)	The built-in potential that opposes the flow of majority carriers across the junction; 0.7 V for Si, 0.3 V for Ge.
Forward bias	Positive voltage applied to the p-region; reduces the barrier potential and allows current flow.
Reverse bias	Positive voltage applied to the n-region; increases the barrier potential and blocks current flow.
Knee voltage	The minimum forward voltage beyond which the forward current rises exponentially.
Rectifier	A circuit that converts AC to DC using diodes.
Clipper	A circuit that removes part of the input waveform without distorting the remainder.
Clamper	A circuit that shifts the DC level of a waveform without changing its shape.
Ripple factor	A measure of the AC content in the DC output of a rectifier; lower is better.
PIV (Peak Inverse Voltage)	The maximum reverse voltage a diode must withstand without breaking down.